

一、选择题 1. C 2. B 3. D 4. A 5. C

二、填空题 1. 3, 3.165 2. (4.412, 5.588) 5.495 3. $\frac{2-\bar{X}}{8}$ $\frac{1}{5}$

4. F(6, 3) t(2) 5. $H_0: p = 5\%$ $H_1: p < 5\%$ 0.1

三、

(1) $E(X) = \int_0^1 \theta x^\theta dx = \frac{\theta}{\theta+1}$ (2分)

令 $\bar{X} = E(X) = \frac{\theta}{\theta+1}$ 得 θ 的矩估计量为 $\frac{\bar{X}}{1-\bar{X}}$ (3分)

(2) $L(\theta) = \prod_{i=1}^n p(x_i; \theta) = \prod_{i=1}^n \theta x_i^{\theta-1}$

$\ln L(\theta) = n \ln \theta + (\theta-1) \sum_{i=1}^n \ln x_i$ (3分)

令 $\frac{\partial \ln L(\theta)}{\partial \theta} = 0$ 得 θ 的极大矩估计量为 $\frac{-n}{\sum_{i=1}^n \ln x_i}$ (2分)

四、解：(1) 设 X_1, X_2 的 $\bar{X} \sim N(0, \frac{\sigma^2}{2})$, 即 $\bar{X} \sim N(0, 1)$,

$$P\left\{|\bar{X}| \leq 1\right\} = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 0.6826$$

(2) $X_1 + X_2 \sim N(0, 2\sigma^2), X_1 - X_2 \sim N(0, 2\sigma^2)$, 得 $\frac{X_1 + X_2}{\sigma\sqrt{2}} \sim N(0, 1), \frac{X_1 - X_2}{\sigma\sqrt{2}} \sim N(0, 1)$,

$$\left(\frac{X_1 + X_2}{\sigma\sqrt{2}}\right)^2 \sim \chi^2(1), \left(\frac{X_1 - X_2}{\sigma\sqrt{2}}\right)^2 \sim \chi^2(1), \quad F = \frac{(X_1 + X_2)^2}{(X_1 - X_2)^2} \sim F(1, 1)$$

由 $F_{0.90}(1, 1) = 39.9$, 故 $P\left\{\frac{(X_1 + X_2)^2}{(X_1 - X_2)^2} \leq 39.9\right\} = 0.9$

$$\begin{aligned}
 & \text{五、解} & H_0: \mu = 21 & H_1: \mu < 21 & (3 \text{ 分}) \\
 & & n=17 & \bar{x}=20 & s^*=3.984 & t_{0.95}(16)=1.7459 & (3 \text{ 分}) \\
 & & T = \frac{\bar{x} - 21}{\frac{s^*}{\sqrt{n}}} & t = -1.0349 > -t_{0.95}(16) & (6 \text{ 分})
 \end{aligned}$$

所以接受原假设，认为这批饮品符合规定。 (3 分)